09. Number Codes

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Simple encoding:

Using the code A=1, B=2, C=3 ... Z=26, what is the code for DOG?

Answer: D=4, O=15, G=7. Answer: **4-15-7**.

Example 2 — Decoding:

Using the code A=1, B=2, C=3 ... Z=26, which word does 12-9-15-14 represent?

Answer: 12=L, 9=I, 15=O, 14=N, spelling LION. Answer: **LION**.

Example 3 — Multiple-choice encoding:

Using the code A=1, B=2, C=3 ... Z=26, what is the code for MILK? A) 13-9-12-11 B) 13-9-12-10 C) 12-9-12-11 D) 13-8-12-11

Answer: M=13, I=9, L=12, K=11, giving 13-9-12-11. Answer: **A) 13-9-12-11**. Check every digit carefully — each wrong option changes just one number.

Example 4 — Multiple-choice decoding:

Using the code A=1, B=2, C=3 ... Z=26, which word does 3-8-5-6 represent? A) CHEF B) CHIEF C) CHEW D) CHIP

Answer: 3=C, 8=H, 5=E, 6=F, spelling CHEF. Answer: **A) CHEF**.

Example 5 — Select-all:

Select ALL of these codes that are correct for the word shown. A) DOG = 4-15-7 B) LION = 12-9-15-13 C) MILK = 13-9-12-11 D) CHEF = 3-8-5-7

Answer: DOG = 4-15-7 (A) and MILK = 13-9-12-11 (C) are both correct. LION should be 12-9-15-14, not 12-9-15-13, so B is wrong; CHEF should end in 6, not 7, so D is wrong. Answer: **A and C**.

Foundation

15 questions

1. Using the code A=1, B=2, C=3 ... Z=26, what is the code for TOP?

2. Using the code A=1, B=2, C=3 ... Z=26, what is the code for RUN?

3. Using the code A=1, B=2, C=3 ... Z=26, what is the code for SKY?

4. Using the code A=1, B=2, C=3 ... Z=26, what is the code for ICE?

5. Using the code A=1, B=2, C=3 ... Z=26, what is the code for OWL?

6. Using the code A=1, B=2, C=3 ... Z=26, which word does 20-5-1 represent? A) TEA B) SEA C) TEN D) TEE

7. Using the code A=1, B=2, C=3 ... Z=26, what is the code for BAT?

8. Using the code A=1, B=2, C=3 ... Z=26, what is the code for FOX?

9. Using the code A=1, B=2, C=3 ... Z=26, what is the code for WIN?

10. Using the code A=1, B=2, C=3 ... Z=26, which word does 8-1-20 represent? A) HAT B) HAS C) HIT D) HAD

11. Using the code A=1, B=2, C=3 ... Z=26, what is the code for VAN?

12. Using the code A=1, B=2, C=3 ... Z=26, what is the code for ZOO?

13. Select ALL the words below whose code would start with a number greater than 20. A) VAN B) TOP C) ZOO D) YAK E) JOY

14. Using the code A=1, B=2, C=3 ... Z=26, what is the code for JOY?

15. Using the code A=1, B=2, C=3 ... Z=26, what is the code for COW?

Standard

20 questions

1. Using the code A=1, B=2, C=3 ... Z=26, what is the code for WOLF?

2. Using the code A=1, B=2, C=3 ... Z=26, what is the code for CRAB?

3. Using the code A=1, B=2, C=3 ... Z=26, what is the code for SWAN?

4. Using the code A=1, B=2, C=3 ... Z=26, which word does 20-9-7-5-18 represent?

5. Using the code A=1, B=2, C=3 ... Z=26, what is the code for CLOUD?

6. Using the code A=1, B=2, C=3 ... Z=26, what is the code for PLANT? A) 16-12-1-14-20 B) 16-12-1-15-20 C) 16-11-1-14-20 D) 16-12-2-14-20

7. Using the code A=1, B=2, C=3 ... Z=26, what is the code for BRICK?

8. Using the code A=1, B=2, C=3 ... Z=26, which word does 3-8-1-9-18 represent? A) CHAIR B) CHASE C) CHART D) CHAIN E) CHIRP

9. Using the code A=1, B=2, C=3 ... Z=26, what is the code for GRAPE?

10. Using the code A=1, B=2, C=3 ... Z=26, what is the code for TRAIN?

11. Select ALL the words below whose code would start with a number greater than 15. A) CANDY B) MANGO C) TULIP D) VIOLA E) SNAIL

12. Using the code A=1, B=2, C=3 ... Z=26, which word does 3-1-13-5-12 represent?

13. Using the code A=1, B=2, C=3 ... Z=26, what is the code for TOWER?

14. Using the code A=1, B=2, C=3 ... Z=26, which word does 7-8-15-19-20 represent? A) GHOST B) GHOSE C) GHOTS D) GHOAT E) GOHST

15. Using the code A=1, B=2, C=3 ... Z=26, what is the code for PEARL?

16. Using the code A=1, B=2, C=3 ... Z=26, what is the code for SHEEP?

17. Using the code A=1, B=2, C=3 ... Z=26, which word does 19-23-1-14 represent?

18. Using the code A=1, B=2, C=3 ... Z=26, what is the code for MANGO?

19. Using the code A=1, B=2, C=3 ... Z=26, what is the code for TULIP?

20. Select ALL of these codes that are correct for the word shown. A) WOLF = 23-15-12-6 B) CRAB = 3-18-1-3 C) GRAPE = 7-18-1-16-5 D) PEARL = 16-5-1-19-12 E) TOWER = 20-15-23-5-18

Hard

5 questions

1. Using the code A=1, B=2, C=3 ... Z=26, what is the code for MOUSE?

2. Using the code A=1, B=2, C=3 ... Z=26, which word does 8-15-21-19-5 represent? A) HOUSE B) HORSE C) MOUSE D) HOUSED

3. Select ALL of these codes that are correct for the word shown. A) CASTLE = 3-1-19-20-12-5 B) FOREST = 6-15-18-5-19-21 C) GARDEN = 7-1-18-4-5-14 D) SPIDER = 19-16-9-4-5-19 E) MOUSE = 13-15-21-19-5

4. Using the code A=1, B=2, C=3 … Z=26, what is the code for FOREST?

5. Using the code A=1, B=2, C=3 … Z=26, what is the code for SPIDER? A) 19-16-9-4-5-18 B) 19-16-9-4-5-19 C) 19-15-9-4-5-18 D) 19-16-9-5-5-18 E) 18-16-9-4-5-18

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